

Supporting Figures

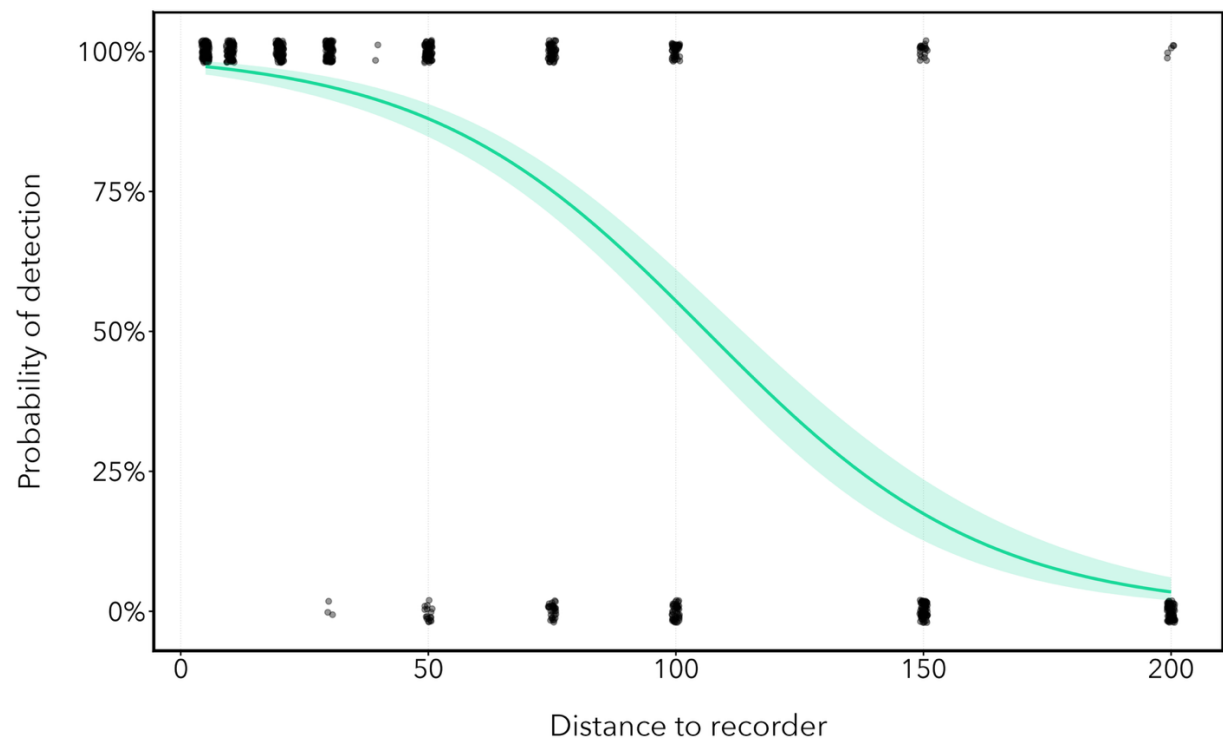


Figure S1. Probability of detecting the call of *Litoria gracilenta* at increasing distance from the recorder, based on field playback trials using the same recording equipment (Frontier Labs Solar BAR with a Primo EM172 microphone). Broadcast calls were played at distances of 5–200 m and scored as detected or not detected in the resulting recordings. The green line shows the fitted probability of detection from a logistic regression, and the shaded band represents the 95% confidence interval. Black points represent individual playback trials (detected = 1, not detected = 0) and are jittered vertically and horizontally to reduce overplotting.

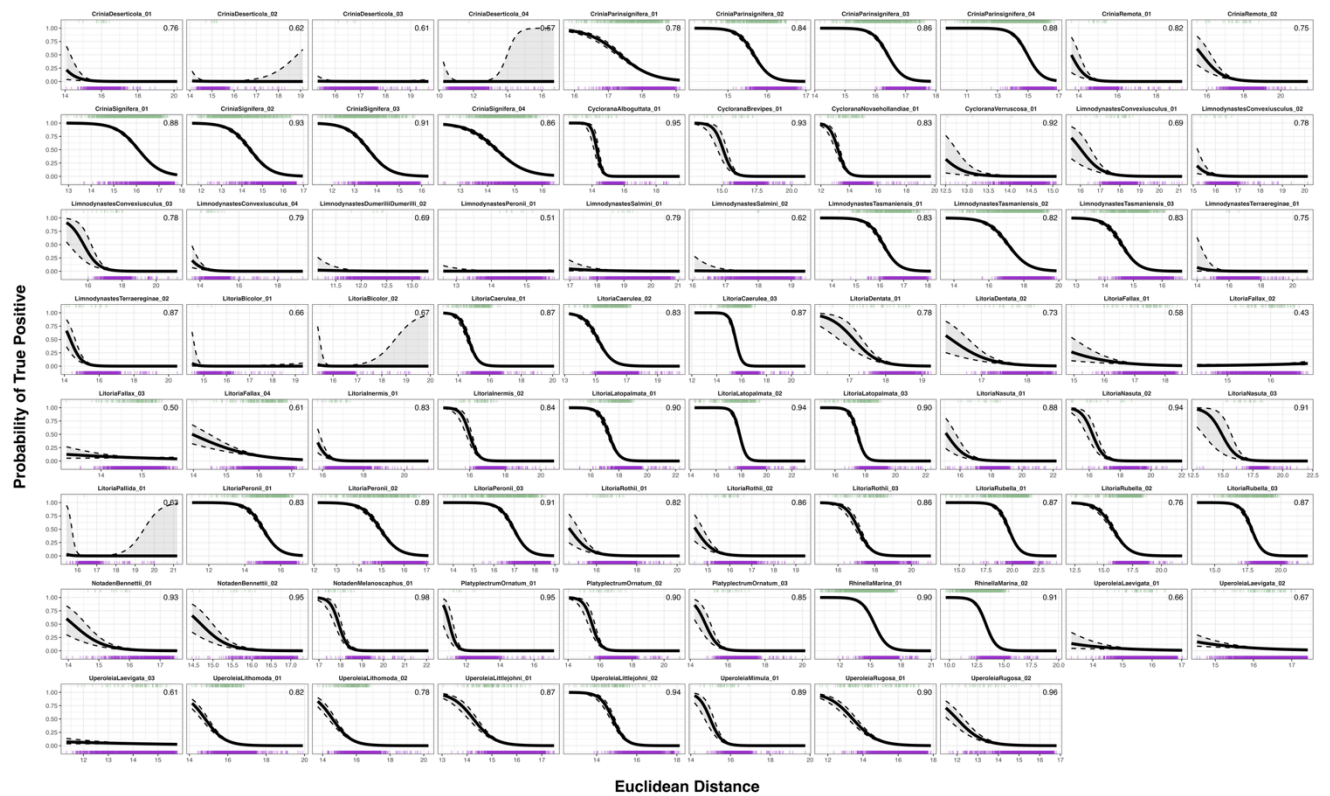


Figure S2. Probability of correctly predicting true positive detections based on the Euclidean distance for each of the 115 example calls, analysed for the 50 frog species used in this study. The green bars represent true positive detections and purple bars false positive detections. The dashed lines and grey area indicate the 95% confidence intervals. The number in the top right corner of each plot indicates the probability that a true positive detection had a lower Euclidean distance than a false positive detection on any given day for that specific example call. Lower Euclidean distances generally corresponded to a higher probability of correctly detecting of an example call.

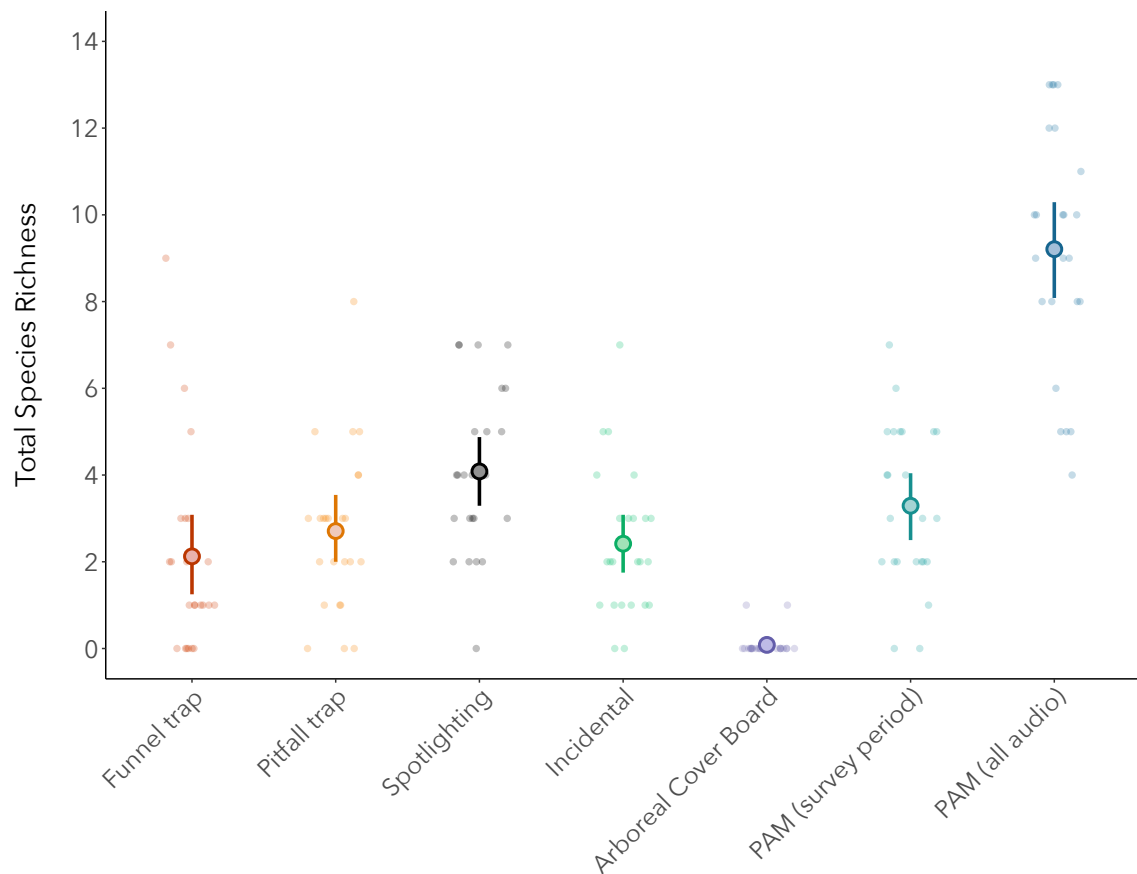
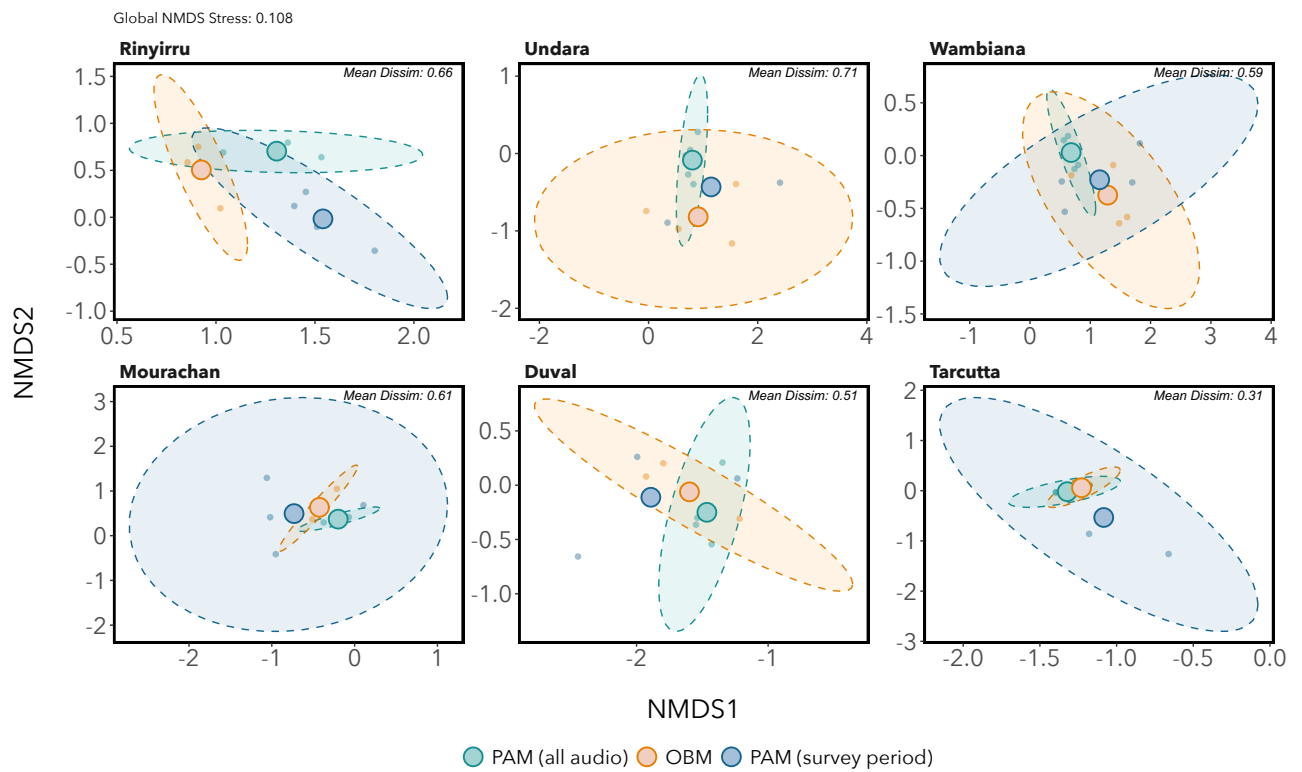


Figure S3. Total species richness of frogs for each survey plot at each site detected by each individual survey method: observer-based monitoring split into funnel trapping (red), pitfall trapping (orange), spotlighting (black), incidental encounters (green), and arboreal cover boards (purple), passive acoustic monitoring using only audio data matching the survey period (light blue), and passive acoustic monitoring using all available audio data (dark blue). Points and error bars represent the mean $\pm$ 95% confidence intervals.



30

31 **Figure S4.** Community composition analysis comparing frog survey methods at each
 32 site separately. Non-metric multidimensional scaling (NMDS) ordination of frog
 33 communities based on Jaccard dissimilarity (presence-absence). Large points
 34 represent the centroids for each method, shaded regions indicate 95% confidence
 35 ellipses, and small points represent individual survey plots.

36 Supporting Tables

37 **Table S1.** Overview of the number and source of the 115 example calls for 50 Australian
38 frog species used in this study.

Common Name	Scientific Name	Example Call	Source
Bufonidae			
Cane Toad	<i>Rhinella marina</i>	RhinellaMarina_01	David Stewart
Cane Toad	<i>Rhinella marina</i>	RhinellaMarina_02	David Stewart
Hylidae			
Striped Burrowing Frog	<i>Cyclorana alboguttata</i>	CycloranaAlboguttata_01	David Stewart
Superb Collared Frog	<i>Cyclorana brevipes</i>	CycloranaBrevipes_01	David Stewart
Knife-footed Frog	<i>Cyclorana cultripes</i>	CycloranaCultripes_01	David Stewart
Knife-footed Frog	<i>Cyclorana cultripes</i>	CycloranaCultripes_02	David Stewart
Eastern Snapping Frog	<i>Cyclorana novaehollandiae</i>	CycloranaNovaehollandiae_01	David Stewart
Eastern Water-holding Frog	<i>Cyclorana platycephalus</i>	CycloranaPlatycephala_01	FrogID
Eastern Water-holding Frog	<i>Cyclorana platycephalus</i>	CycloranaPlatycephala_02	FrogID
Rough Collared Frog	<i>Cyclorana verrucosa</i>	CycloranaVerrucosa_01	David Stewart
Northern Sedge Frog	<i>Carichyla bicolor</i>	LitoriaBicolor_01	David Stewart
Northern Sedge Frog	<i>Carichyla bicolor</i>	LitoriaBicolor_02	David Stewart
Booroolong Frog	<i>Rhyaconastes booroolongensis</i>	LitoriaBooroolongensis_01	Michael Mahony
Booroolong Frog	<i>Rhyaconastes booroolongensis</i>	LitoriaBooroolongensis_02	Michael Mahony
Booroolong Frog	<i>Rhyaconastes booroolongensis</i>	LitoriaBooroolongensis_03	Michael Mahony
Green Tree Frog	<i>Pelodryas caerulea</i>	LitoriaCaerulea_01	David Stewart
Green Tree Frog	<i>Pelodryas caerulea</i>	LitoriaCaerulea_02	David Stewart
Green Tree Frog	<i>Pelodryas caerulea</i>	LitoriaCaerulea_03	David Stewart
Robust Bleating Tree Frog	<i>Coleeneremia dentata</i>	LitoriaDentata_01	David Stewart
Robust Bleating Tree Frog	<i>Coleeneremia dentata</i>	LitoriaDentata_02	David Stewart
Brown Tree Frog	<i>Rawlinsonia ewingii</i>	LitoriaEwingii_01	FrogID
Brown Tree Frog	<i>Rawlinsonia ewingii</i>	LitoriaEwingii_02	FrogID
Brown Tree Frog	<i>Rawlinsonia ewingii</i>	LitoriaEwingii_03	FrogID
Eastern Dwarf Tree Frog	<i>Drymomantis fallax</i>	LitoriaFallax_01	David Stewart
Eastern Dwarf Tree Frog	<i>Drymomantis fallax</i>	LitoriaFallax_02	David Stewart
Eastern Dwarf Tree Frog	<i>Drymomantis fallax</i>	LitoriaFallax_03	David Stewart
Eastern Dwarf Tree Frog	<i>Drymomantis fallax</i>	LitoriaFallax_04	David Stewart
Bumpy Rocket Frog	<i>Litoria inermis</i>	LitoriaInermis_01	David Stewart
Bumpy Rocket Frog	<i>Litoria inermis</i>	LitoriaInermis_02	David Stewart
Broad-palmed Rocket Frog	<i>Litoria latopalmata</i>	LitoriaLatopalmata_01	David Stewart
Broad-palmed Rocket Frog	<i>Litoria latopalmata</i>	LitoriaLatopalmata_02	David Stewart
Broad-palmed Rocket Frog	<i>Litoria latopalmata</i>	LitoriaLatopalmata_03	David Stewart
Javelin Frog	<i>Mahonabatrachus microbelos</i>	LitoriaMicrobelos_01	David Stewart
Javelin Frog	<i>Mahonabatrachus microbelos</i>	LitoriaMicrobelos_02	David Stewart
Striped Rocket Frog	<i>Litoria nasuta</i>	LitoriaNasuta_01	David Stewart
Striped Rocket Frog	<i>Litoria nasuta</i>	LitoriaNasuta_02	David Stewart
Striped Rocket Frog	<i>Litoria nasuta</i>	LitoriaNasuta_03	David Stewart
Tawney Rocket Frog	<i>Litoria nigrofrenata</i>	LitoriaNigrofrenata_01	David Stewart
Pale Rocket Frog	<i>Litoria pallida</i>	LitoriaPallida_01	David Stewart
Pale Rocket Frog	<i>Litoria pallida</i>	LitoriaPallida_02	David Stewart
Peron's Tree Frog	<i>Pengilleyia peronii</i>	LitoriaPeronii_01	David Stewart
Peron's Tree Frog	<i>Pengilleyia peronii</i>	LitoriaPeronii_02	David Stewart
Peron's Tree Frog	<i>Pengilleyia peronii</i>	LitoriaPeronii_03	David Stewart
Southern Bell Frog	<i>Ranoidea raniformis</i>	LitoriaRaniformis_01	Sarah Talbot
Southern Bell Frog	<i>Ranoidea raniformis</i>	LitoriaRaniformis_02	Sarah Talbot
Northern Laughing Tree Frog	<i>Pengilleyia rothii</i>	LitoriaRothii_01	David Stewart
Northern Laughing Tree Frog	<i>Pengilleyia rothii</i>	LitoriaRothii_02	David Stewart
Northern Laughing Tree Frog	<i>Pengilleyia rothii</i>	LitoriaRothii_03	David Stewart
Ruddy Tree Frog	<i>Coleeneremia pyrina</i>	LitoriaRubella_01	David Stewart
Ruddy Tree Frog	<i>Coleeneremia pyrina</i>	LitoriaRubella_02	David Stewart
Ruddy Tree Frog	<i>Coleeneremia pyrina</i>	LitoriaRubella_03	David Stewart
Tyler's Tree Frog	<i>Pengilleyia tyleri</i>	LitoriaTyleri_01	David Stewart
Tyler's Tree Frog	<i>Pengilleyia tyleri</i>	LitoriaTyleri_02	David Stewart
Whistling Tree Frog	<i>Rawlinsonia verreauxii verreauxii</i>	LitoriaVerreauxiiVereauxii_01	David Stewart
Whistling Tree Frog	<i>Rawlinsonia verreauxii verreauxii</i>	LitoriaVerreauxiiVereauxii_02	David Stewart

Common Name	Scientific Name	Example Call	Source
Limnodynastidae			
Tusked Frog	<i>Adelotus brevis</i>	AdelotusBrevis_01	David Stewart
Tusked Frog	<i>Adelotus brevis</i>	AdelotusBrevis_02	David Stewart
Tusked Frog	<i>Adelotus brevis</i>	AdelotusBrevis_03	David Stewart
Tusked Frog	<i>Adelotus brevis</i>	AdelotusBrevis_04	David Stewart
Marbled Frog	<i>Limnodynastes convexiusculus</i>	LimnodynastesConvexiusculus_01	David Stewart
Marbled Frog	<i>Limnodynastes convexiusculus</i>	LimnodynastesConvexiusculus_02	David Stewart
Marbled Frog	<i>Limnodynastes convexiusculus</i>	LimnodynastesConvexiusculus_03	David Stewart
Marbled Frog	<i>Limnodynastes convexiusculus</i>	LimnodynastesConvexiusculus_04	David Stewart
Eastern Banjo Frog	<i>Limnodynastes dumerilii dumerilii</i>	LimnodynastesDumeriliiDumerilli_01	David Stewart
Eastern Banjo Frog	<i>Limnodynastes dumerilii dumerilii</i>	LimnodynastesDumeriliiDumerilli_02	David Stewart
Eastern Banjo Frog	<i>Limnodynastes dumerilii dumerilii</i>	LimnodynastesDumeriliiDumerilli_03	David Stewart
Barking Frog	<i>Limnodynastes fletcheri</i>	LimnodynastesFletcheri_01	David Stewart
Giant Banjo Frog	<i>Limnodynastes interioris</i>	LimnodynastesInterioris_01	Sarah Talbot
Giant Banjo Frog	<i>Limnodynastes interioris</i>	LimnodynastesInterioris_02	Sarah Talbot
Giant Banjo Frog	<i>Limnodynastes interioris</i>	LimnodynastesInterioris_03	Sarah Talbot
Striped Marsh Frog	<i>Limnodynastes peronii</i>	LimnodynastesPeronii_01	David Stewart
Striped Marsh Frog	<i>Limnodynastes peronii</i>	LimnodynastesPeronii_02	David Stewart
Salmon-striped Frog	<i>Limnodynastes salmini</i>	LimnodynastesSalmini_01	David Stewart
Salmon-striped Frog	<i>Limnodynastes salmini</i>	LimnodynastesSalmini_02	David Stewart
Spotted Marsh Frog	<i>Limnodynastes tasmaniensis</i>	LimnodynastesTasmaniensis_01	David Stewart
Spotted Marsh Frog	<i>Limnodynastes tasmaniensis</i>	LimnodynastesTasmaniensis_02	David Stewart
Spotted Marsh Frog	<i>Limnodynastes tasmaniensis</i>	LimnodynastesTasmaniensis_03	David Stewart
Superb Banjo Frog	<i>Limnodynastes terraereginae</i>	LimnodynastesTerraereginae_01	David Stewart
Superb Banjo Frog	<i>Limnodynastes terraereginae</i>	LimnodynastesTerraereginae_02	David Stewart
Sudell's Frog	<i>Neobatrachus sudellae</i>	NeobatrachisSudelli_01	Dale Roberts
Holy Cross Frog	<i>Notaden bennettii</i>	NotadenBennettii_01	David Stewart
Holy Cross Frog	<i>Notaden bennettii</i>	NotadenBennettii_02	David Stewart
Northern Spadefoot	<i>Notaden melanoscaphus</i>	NotadenMelanoscapus_01	David Stewart
Ornate Burrowing Frog	<i>Platyplectrum ornatum</i>	PlatyplectrumOrnatum_01	David Stewart
Ornate Burrowing Frog	<i>Platyplectrum ornatum</i>	PlatyplectrumOrnatum_02	David Stewart
Ornate Burrowing Frog	<i>Platyplectrum ornatum</i>	PlatyplectrumOrnatum_03	David Stewart
Myobatrachidae			
Desert Froglet	<i>Crinia deserticola</i>	CriniaDeserticola_01	David Stewart
Desert Froglet	<i>Crinia deserticola</i>	CriniaDeserticola_02	David Stewart
Desert Froglet	<i>Crinia deserticola</i>	CriniaDeserticola_03	David Stewart
Desert Froglet	<i>Crinia deserticola</i>	CriniaDeserticola_04	David Stewart
Beeping Froglet	<i>Crinia parinsignifera</i>	CriniaParinsignifera_01	David Stewart
Beeping Froglet	<i>Crinia parinsignifera</i>	CriniaParinsignifera_02	David Stewart
Beeping Froglet	<i>Crinia parinsignifera</i>	CriniaParinsignifera_03	David Stewart
Beeping Froglet	<i>Crinia parinsignifera</i>	CriniaParinsignifera_04	David Stewart
Northern Froglet	<i>Crinia remota</i>	CriniaRemota_01	David Stewart
Northern Froglet	<i>Crinia remota</i>	CriniaRemota_02	David Stewart
Clicking Froglet	<i>Crinia signifera</i>	CriniaSignifera_01	David Stewart
Clicking Froglet	<i>Crinia signifera</i>	CriniaSignifera_02	David Stewart
Clicking Froglet	<i>Crinia signifera</i>	CriniaSignifera_03	David Stewart
Clicking Froglet	<i>Crinia signifera</i>	CriniaSignifera_04	David Stewart
Sloane's Froglet	<i>Crinia sloanei</i>	CriniaSloanei_01	Alexandra Knight
Sloane's Froglet	<i>Crinia sloanei</i>	CriniaSloanei_02	Alexandra Knight
Sloane's Froglet	<i>Crinia sloanei</i>	CriniaSloanei_02	Alexandra Knight
Brown Toadlet	<i>Pseudophryne bibronii</i>	PseudophryneBibronii_01	David Stewart
Brown Toadlet	<i>Pseudophryne bibronii</i>	PseudophryneBibronii_02	David Stewart
Smooth Toadlet	<i>Uperoleia laevigata</i>	UperoleiaLaevigata_01	David Stewart
Smooth Toadlet	<i>Uperoleia laevigata</i>	UperoleiaLaevigata_02	David Stewart
Smooth Toadlet	<i>Uperoleia laevigata</i>	UperoleiaLaevigata_03	David Stewart
Stonemason Toadlet	<i>Uperoleia lithomoda</i>	UperoleiaLithomoda_01	David Stewart
Stonemason Toadlet	<i>Uperoleia lithomoda</i>	UperoleiaLithomoda_02	David Stewart
Einasteigh Toadlet	<i>Uperoleia littlejohni</i>	UperoleiaLittlejohni_01	David Stewart
Einasteigh Toadlet	<i>Uperoleia littlejohni</i>	UperoleiaLittlejohni_02	David Stewart
Mimic Toadlet	<i>Uperoleia mimula</i>	UperoleiaMimula_01	David Stewart
Chubby Toadlet	<i>Uperoleia rugosa</i>	UperoleiaRugosa_01	David Stewart
Chubby Toadlet	<i>Uperoleia rugosa</i>	UperoleiaRugosa_02	David Stewart

40 **Table S2.** Detailed summary of the survey dates, acoustic recording hours, and camera trapping days (summed for all four camera traps) at a survey
41 plot level across the six study sites.

Site	Plot	First survey			Second survey			Third survey			Fourth survey			Total survey days	Total camera trapping days	Total acoustic data (survey period) in hours	Total acoustic data (all audio) in hours
		Dates	Acoustic data (hours)	Camera trapping days	Dates	Acoustic data (hours)	Camera trapping days	Dates	Acoustic data (hours)	Camera trapping days	Dates	Acoustic data (hours)	Camera trapping days				
Tarcutta	Dry A	29/4/2021 – 6/5/2021	167	28	18/10/2021 – 25/10/2021	168	14	8/5/2022 – 15/5/2022	154	28	22/11/2022 – 29/11/2022	160	28	28	98	649	18600
Tarcutta	Dry B	29/4/2021 – 6/5/2021	167	28	18/10/2021 – 25/10/2021	168	28	8/5/2022 – 15/5/2022	166	28	22/11/2022 – 29/11/2022	158	28	28	112	659	22356
Tarcutta	Wet A	29/4/2021 – 6/5/2021	163	28	18/10/2021 – 25/10/2021	168	28	8/5/2022 – 15/5/2022	164	28	22/11/2022 – 29/11/2022	158	28	28	112	653	8605
Tarcutta	Wet B	29/4/2021 – 6/5/2021	167	28	18/10/2021 – 25/10/2021	168	28	8/5/2022 – 15/5/2022	163	28	22/11/2022 – 29/11/2022	158	28	28	112	656	20289
Duval	Dry A	18/4/2021 – 25/4/2021	167	28	NA	NA	NA	28/4/2022 – 5/5/2022	166	28	12/11/2022 – 19/11/2022	129	28	21	84	462	11397
Duval	Dry B	18/4/2021 – 25/4/2021	168	28	NA	NA	NA	NA	NA	NA	12/11/2022 – 19/11/2022	144	28	14	56	312	16209
Duval	Wet A	18/4/2021 – 25/4/2021	167	28	NA	NA	NA	28/4/2022 – 5/5/2022	163	28	12/11/2022 – 19/11/2022	148	28	21	84	478	12402
Duval	Wet B	18/4/2021 – 25/4/2021	168	21	NA	NA	NA	NA	NA	NA	12/11/2022 – 19/11/2022	133	28	14	49	301	10921
Mourachan	Dry A	9/5/2021 – 16/5/2021	167	28	NA	NA	NA	19/6/2022 – 26/6/2022	167	28	2/11/2022 – 9/11/2022	167	28	21	84	501	14894
Mourachan	Dry B	9/5/2021 – 16/5/2021	8	28	NA	NA	NA	19/6/2022 – 26/6/2022	1	28	2/11/2022 – 9/11/2022	166	28	21	84	175	10667
Mourachan	Wet A	9/5/2021 – 16/5/2021	118	28	NA	NA	NA	19/6/2022 – 26/6/2022	4	28	2/11/2022 – 9/11/2022	166	28	21	84	288	13299

Site	Plot	First survey			Second survey			Third survey			Fourth survey			Total survey days	Total camera trapping days	Total acoustic data (survey period) in hours	Total acoustic data (all audio) in hours
		Dates	Acoustic data (hours)	Camera trapping days	Dates	Acoustic data (hours)	Camera trapping days	Dates	Acoustic data (hours)	Camera trapping days	Dates	Acoustic data (hours)	Camera trapping days				
Mourachan	Wet B	9/5/2021 – 16/5/2021	158	28	NA	NA	NA	19/6/2022 – 26/6/2022	165	28	2/11/2022 – 9/11/2022	166	28	21	84	489	8083
Wambiana	Dry A	5/7/2021 – 12/7/2021	165	28	10/11/2021 – 17/11/2021	16	28	12/6/2022 – 19/6/2022	165	28	28/9/2022 – 5/10/2022	166	28	28	112	512	27108
Wambiana	Dry B	5/7/2021 – 12/7/2021	164	28	10/11/2021 – 17/11/2021	163	28	12/6/2022 – 19/6/2022	166	28	28/9/2022 – 5/10/2022	3	28	28	112	496	15323
Wambiana	Wet A	5/7/2021 – 12/7/2021	166	28	10/11/2021 – 17/11/2021	166	28	12/6/2022 – 19/6/2022	164	28	28/9/2022 – 5/10/2022	166	28	28	112	662	15108
Wambiana	Wet B	5/7/2021 – 12/7/2021	164	28	10/11/2021 – 17/11/2021	158	28	12/6/2022 – 19/6/2022	161	28	28/9/2022 – 5/10/2022	167	28	28	112	650	21185
Undara	Dry A	3/6/2021 – 10/6/2021	164	28	29/9/2021 – 6/10/2021	164	28	8/5/2022 – 15/5/2022	146	28	13/10/2022 – 20/10/2022	160	28	28	112	634	11971
Undara	Dry B	3/6/2021 – 10/6/2021	161	28	29/9/2021 – 6/10/2021	155	28	8/5/2022 – 15/5/2022	151	28	13/10/2022 – 20/10/2022	163	28	28	112	630	12280
Undara	Wet A	3/6/2021 – 10/6/2021	159	28	29/9/2021 – 6/10/2021	41	28	8/5/2022 – 15/5/2022	NA	28	13/10/2022 – 20/10/2022	NA	28	27	112	200	4382
Undara	Wet B	3/6/2021 – 10/6/2021	165	28	29/9/2021 – 6/10/2021	95	28	8/5/2022 – 15/5/2022	133	28	13/10/2022 – 20/10/2022	158	28	27	112	551	11212
Rinyirru	Dry A	14/6/2021 – 21/6/2021	166	21	8/10/2021 – 15/10/2021	145	28	7/8/2022 – 14/8/2022	137	28	23/10/2022 – 30/10/2022	130	28	28	105	578	8548
Rinyirru	Dry B	14/6/2021 – 21/6/2021	159	28	8/10/2021 – 15/10/2021	110	28	7/8/2022 – 14/8/2022	161	28	23/10/2022 – 30/10/2022	160	28	28	112	590	10495
Rinyirru	Wet A	14/6/2021 – 21/6/2021	158	28	8/10/2021 – 15/10/2021	142	28	7/8/2022 – 14/8/2022	146	28	23/10/2022 – 30/10/2022	135	28	28	112	581	5898
Rinyirru	Wet B	14/6/2021 – 21/6/2021	161	28	8/10/2021 – 15/10/2021	144	28	7/8/2022 – 14/8/2022	141	28	23/10/2022 – 30/10/2022	150	28	28	112	596	6178

Table S3. Frog species detected via observer-based monitoring (OBM) through pitfall traps (PF), funnel traps (FT), arboreal cover boards (ACB), spotlighting (SL), and incidental encounters (IE); and passive acoustic monitoring (PAM) for both matching only the survey period, and all available audio data. The numbers for each species represent instances of detection rather than abundances.

Common name	Scientific name	Observer-based monitoring (OBM)					Passive acoustic monitoring (PAM)	
		PT	FT	ACB	SL	IE	Survey period	All audio
Bufonidae								
Cane Toad	<i>Rhinella marina</i>	143	119		321	19	65	1547
Hylidae								
Striped Burrowing Frog	<i>Cyclorana alboguttata</i>							33
Superb Collared Frog	<i>Cyclorana brevipes</i>							25
Eastern Snapping Frog	<i>Cyclorana novaehollandiae</i>				8	8	3	67
Rough Collared Frog	<i>Cyclorana verrucosa</i>		2					3
Northern Sedge Frog	<i>Carichyla bicolor</i>							3
Green Tree Frog	<i>Pelodryas caerulea</i>	4	2		52	19	35	717
Robust Bleating Tree Frog	<i>Coleeneremia dentata</i>							56
Eastern Dwarf Tree Frog	<i>Drymomantis fallax</i>				82	4	19	446
Bumpy Rocket Frog	<i>Litoria inermis</i>	1	9		40		1	76
Broad-palmed Rocket Frog	<i>Litoria latopalmata</i>	4	7		187	27	50	582
Striped Rocket Frog	<i>Litoria nasuta</i>	8	4		1	1	1	57
Pale Rocket Frog	<i>Litoria pallida</i>							1
Peron's Tree Frog	<i>Pengilleyia peronii</i>				276	10	54	1164
Northern Laughing Tree Frog	<i>Pengilleyia rothii</i>					1	1	184
Little Red Tree Frog	<i>Coleeneremia pyrina</i>	26	6	2	107	16	26	1108
Limnodynastidae								
Marbled Frog	<i>Limnodynastes convexiusculus</i>	4	1				1	75
Eastern Banjo Frog	<i>Limnodynastes dumerilii dumerilii</i>	1			33	11	1	5
Scarlet-sided Banjo Frog	<i>Limnodynastes grayi</i>	6	4		5	2	8	21
Striped Marsh Frog	<i>Limnodynastes peronii</i>	1						8
Salmon-Striped Frog	<i>Limnodynastes salmini</i>							11
Spotted Marsh Frog	<i>Limnodynastes tasmaniensis</i>	54	10		160	9	53	1413
Superb Banjo Frog	<i>Limnodynastes terraereginae</i>	2	3					
Holy Cross Frog	<i>Notaden bennettii</i>	3			5	1		28
Northern Spadefoot	<i>Notaden melanoscapus</i>	23	4		15	2		26
Ornate Burrowing Frog	<i>Platyplectrum ornatum</i>	95	22	1	4	6		65
Myobatrachidae								
Desert Froglet	<i>Crinia deserticola</i>						2	26
Beeping Froglet	<i>Crinia parinsignifera</i>	11	8		603	65	86	2298
Northern Froglet	<i>Crinia remota</i>				1		9	44
Clicking Froglet	<i>Crinia signifera</i>	20	7		208	62	92	2612
Smooth Toadlet	<i>Uperoleia laevigata</i>							166
Stonemason Toadlet	<i>Uperoleia lithomoda</i>						17	323
Einasleigh Toadlet	<i>Uperoleia littlejohni</i>						3	210
Mimic Toadlet	<i>Uperoleia mimula</i>							33
Chubby Toadlet	<i>Uperoleia rugosa</i>	13	2		47		13	144
Total number of species		18	16	2	19	17	21	34

Table S4. Total frog species richness detected via observer-based monitoring (OBM), passive acoustic monitoring (PAM) for both matching only the survey period, and all available audio data, and all methods combined.

	Total species richness			
	OBM	PAM (survey period)	PAM (all audio)	All methods combined
Rinyirru	12	8	16	19
Undara	9	10	15	15
Wambiana	7	6	12	12
Mourachan	9	8	13	13
Duval	6	7	10	10
Tarcutta	6	5	6	6

Table S5. Number of PAM sampling days needed to match the maximum OBM species richness at each site. PAM spring = sampling only in spring (Sep - Dec), PAM summer = sampling only in summer (Dec - Mar), PAM autumn = sampling only in autumn (Mar - Jun), PAM winter = sampling only in winter (Jun - Sep), PAM all season = sampling throughout the whole year.

	Days to match OBM species richness				
	PAM spring	PAM summer	PAM autumn	PAM winter	PAM all season
Rinyirru	–	37	–	–	68
Undara	62	13	221	–	35
Wambiana	35	9	35	61	19
Mourachan	24	39	–	–	40
Duval	4	5	34	51	10
Tarcutta	–	–	–	–	387